

Some tips on effective paper writing

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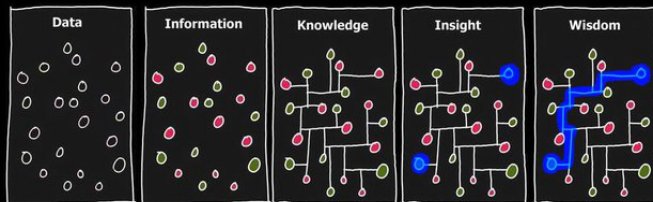
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- How to seduce your audience (objectives of your paper)
- Detailed structure and contents of a paper
- How to avoid plagiarism
- Case studies
-

Objective of your research paper

Show the results of your research before the scientific community.



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We need a logic path

Information → Knowledge → Critical discussion

Information: Introduce your research to your audience.

Paper Title

Short, Informative

Keywords

This is how others find your work

Your research
at a glance

This is how they know whether your work
is what they are looking for

Your motivations

Why you did this?

The state of the art

What has been done before your work?

Your approach

Summarize what you did.
WHAT IS THE NOVELTY

Its advantages and
trade-offs

Why is your research worth? What is better?
What is the price of applying your approach?

Information: Introduce your research to your audience.

Paper Title

Keywords

Your research
at a glance

← This is your abstract

Your motivations

References go here

The state of the art

Your approach

← This is your Introduction

Its advantages and
trade-offs

State of the art

It is not written:

- To show off your wide knowledge of the topic.
- To meaninglessly list all what it is done in the area.

It is important for:

- Putting your reader in context.
- Summarize the methods, advantages and trade-offs of related work.

Summary of your research

- It must clearly explain what you intend to do.
- It must be compared to the above state of the art to **clearly show the novelty of your approaches.**

Knowledge: Introduce your approach

Summarize the theoretical background

Here you introduce what is needed to understand the theory below

Derive your approach

Explain your approach from a theory perspective

Explain your algorithms

Here you explain how to apply your approach

Provide some explanation of the advantages of your approach

Justify its advantages

Knowledge: show the practical application of your approach

What experiments
were conducted?

Here the reader will
learn the big picture of
the experimental part...

What do you want to
prove with them?

... then the reader will
understand why you
designed them as you
described

How did you obtain
the data?

First step of
reproducibility

Detail the experimental
setup

Second step of
reproducibility

Show the results

Here you **MUST** provide
informative representations

Fully describe your experiments

- **What** you did
- **Why** you did it like that

Make sure they are REPRODUCIBLE

- How you generated your data
- All other details: chosen parameters, statistical measures...

Be informative

- Present the results in a **clean** way.
- Present the results in a **clear** way.
- Comment all graphs, figures, etc in your text.
- **Always** use captions.

Discussion

- Analyze your results being **critical** with yourself
- **Compare** your results with the state of the art
- Discuss the **advantages and trade-offs** of your approach.

Conclusion

The conclusion is useful for:

- Reviewing all what you present for your reader to get the big picture
- Mentioning what research paths can be taken from this

So, summarize everything explaining in a structured way what is important, what is worth to remember. Explain your future plans regarding this research, if any.

My conclusion

The purpose of articles is to disseminate knowledge in a useful way. Thus, usually the next sections are needed:

- Introduction: to put your work in context; summarize the state of the art in order to show that your work is original and it addresses problems from new points of view.
- Theory: to explain your approaches with detail. You may need a background summary prior to that.
- Experiments: essential in many papers. The description must be sufficient to reproduce exactly what you did.
- Results must clearly show your claims. Be extremely careful with graphs and other representations that may become messy.
- Discuss in a critical way. Compare yourself in a fair way. Point out what is good and bad of your approach.
- Summarize all so the reader can see the whole thing at a glance.